Game of Stars

Microfoundations of the J-Curve in Online Ratings

Marco Canal Gal·la Gelpí Eric Gutiérrez

Barcelona School of Economics

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Abstract

Online product ratings exhibit a J-shaped distribution: a dominant peak at five stars, a visible tail at one star, and a hollow middle. This pattern persists across platforms, categories, and time, yet product quality is plausibly bell-shaped. We provide a microfounded explanation for this puzzle. Our key insight is that ratings measure calibration, whether a product met price-implied expectations, rather than absolute quality. A cheap product that works as expected and a premium product that works as expected both receive five stars; a premium product that disappoints receives one star. We show that the J-curve emerges statically from asymmetric rating bins: all positive experience-expectation gaps map to a single bin (five stars), while negative gaps are distributed across four bins. This asymmetry, combined with symmetric gaps around zero at market entry, mechanically produces J-shaped ratings from the very first review. We further show that ratings reveal mispricing rather than quality directly, but that price and average rating jointly identify quality via $Q = \tilde{p} + \phi(\bar{r} - 5)$, thereby solving the Akerlof adverse selection problem. In the long run, the J-curve sharpens as correctly priced products and bargains accumulate reviews while lemons stagnate. Match heterogeneity ensures that even successful products retain a one-star tail indefinitely.

Keywords: Online ratings, J-curve, reference dependence, adverse selection, prospect theory

JEL Codes: D82, D83, D91, L15

1 Introduction

The distribution of consumer ratings on online platforms exhibits a striking empirical regularity. Across major e-commerce platforms, approximately 60 to 70 percent of reviews award the maximum rating of five stars, 10 to 15 percent award the minimum rating of one star, and intermediate ratings are comparatively rare. This "J-shaped" distribution is remarkably stable, persisting across product categories, platforms, time periods, and geographies.

The ubiquity of the J-curve presents a puzzle for standard models of quality revelation. If ratings serve as noisy signals of product quality, and if quality is approximately normally distributed across products, as standard economic reasoning would suggest, then the distribution of ratings should itself be approximately normal. Instead, we observe a distribution that appears bimodal, with a dominant mode at the maximum feasible value.

This paper develops a model that explains the J-curve as an equilibrium outcome of rational consumer behavior. Our central theoretical contribution is to reframe what ratings measure. We argue that ratings reflect calibration: the correspondence between realized consumption utility and prior expectations, rather than absolute product quality. When a consumer assigns a rating, she is not answering the question "how good is this product?" but rather "did this product meet my expectations?". Under this interpretation, a low-quality product sold at an appropriately low price receives five stars if it performs as anticipated, while a high-quality product sold at a premium receives one star if it disappoints relative to the expectations that premium induced.

This reframing yields three results that together explain the J-curve.

First, we establish that ratings reveal mispricing rather than quality. Because ratings measure the gap between experience and expectations, a product's average rating indicates whether it is correctly priced, not whether it is intrinsically superior. High-quality products with inflated prices (lemons in the classic sense) receive poor ratings, while low-quality products with commensurate prices receive favorable ratings. Ratings alone therefore do not resolve adverse selection. However, we demonstrate that price and average rating jointly identify quality: consumers can recover quality from the mapping $Q = \tilde{p} + \phi(\bar{r} - 5)$, where $\tilde{p}$ denotes the posted price and $\bar{r}$ the average rating. The Akerlof problem is thus solved not through direct quality revelation but through calibration verification.

Second, we show that the J-curve emerges from an asymmetry in the rating technology, not from an asymmetry in the underlying distribution of consumer experiences. At the point of market entry, when no reviews exist to inform expectations, the distribution of gaps between realized experience and prior expectations is symmetric around zero. The J-curve arises because the mapping from continuous gaps to discrete ratings is asymmetric: all non-negative gaps (cases in which the product met or exceeded expectations) are assigned to a single bin (five stars), while negative gaps are distributed across the remaining four bins. This asymmetric binning mechanically transforms a symmetric input distribution into a J-shaped output distribution.

Third, we characterize how the J-curve evolves dynamically. In the long run, correctly priced products and underpriced products (bargains) accumulate reviews, as favorable ratings attract

subsequent demand. Overpriced products (lemons) stagnate, as unfavorable ratings repel demand. The aggregate distribution of experience-expectation gaps therefore shifts rightward over time as successful products come to dominate review volume. However, idiosyncratic variation in consumer-product match quality ensures that even successful products occasionally disappoint individual consumers, thereby maintaining a positive mass at low ratings indefinitely. The J-curve sharpens but does not degenerate to a point mass.

This paper contributes to several literatures. We build on the foundational work on adverse selection by Akerlof (1970), demonstrating how online rating systems can ameliorate the lemons problem through a mechanism distinct from direct quality certification. Our model of expectation formation draws on the literature on reference-dependent preferences, particularly Kőszegi and Rabin (2006), while our treatment of the posting decision incorporates insights from prospect theory (Kahneman and Tversky, 1979). The premise that extreme experiences are disproportionately salient connects our work to the salience literature (Bordalo et al., 2012, 2013).

The remainder of the paper proceeds as follows. Section 2 reviews related literature. Section 3 presents the model environment. Sections 4 through 7 develop the four stages of the model: information acquisition and reference point formation, demand determination and product choice, experience realization and review posting, and aggregation. Section 8 characterizes the static J-curve. Section 9 analyzes the dynamic evolution of the rating distribution. Section 10 derives testable empirical predictions. Section 11 presents empirical evidence. Section 12 concludes.

2 Literature Review

Online ratings are commonly understood as a market solution to the adverse selection problem identified by Akerlof (1970). Under this view, ratings aggregate private information about quality and transmit it to prospective buyers, thereby sustaining trade in markets that might otherwise unravel. Our model qualifies this interpretation. Ratings do facilitate quality inference, but indirectly: they reveal whether a product is correctly priced, not whether it is high quality. The distinction is consequential. A product with a 4.8-star average may be low quality but appropriately priced, while a product with a 3.2-star average may be high quality but overpriced. Quality recovery requires observing both ratings and prices.

Our model of consumer satisfaction builds on the reference-dependent preference framework of Kőszegi and Rabin (2006), in which utility depends on the gap between realized consumption and a reference point given by prior expectations. We depart from Kőszegi and Rabin in allowing the reference point to incorporate market signals: consumers form expectations based on both posted prices and accumulated ratings, with the weight on ratings increasing in review volume. This formulation captures the intuition that a new product is judged primarily against its price, while an established product is judged primarily against its reputation. The shift in reference point formation as information accumulates is central to our dynamic results.

The posting decision in our model reflects two well-documented features of prospect-theoretic pref-

erences (Kahneman and Tversky, 1979): loss aversion, which implies that disappointments generate stronger affective responses than equivalent positive surprises, and diminishing sensitivity, which implies that marginal responses attenuate with distance from the reference point. We combine these features with a salience-based posting threshold in the spirit of Bordalo et al. (2012, 2013): consumers post reviews when their experience deviates sufficiently from expectations, and remain silent otherwise. This selective posting generates systematic underrepresentation of moderate experiences relative to extreme ones, a pattern documented empirically by Hu et al. (2009). Our contribution is to show that this selection mechanism, combined with asymmetric binning of continuous satisfaction into discrete ratings, is sufficient to generate the J-curve even when underlying quality and match distributions are symmetric.

3 Model Environment

We model the demand side of an online marketplace with heterogeneous products and sequentially arriving consumers. We do not model the supply side: prices and qualities are fixed, sellers are passive, and there are no markups (perfect competition). The platform is a passive aggregator that displays average ratings and review counts.

3.1 Players

A set of products $i \in \{1, \ldots, I\}$ are available. Products are passive and do not take strategic actions. Each product has fixed characteristics (quality, price) determined at market entry. A continuum of consumers indexed by j arrive sequentially at discrete times $t = 0, 1, 2, \ldots$. Each consumer purchases at most one product and may post at most one review. The platform aggregates reviews and displays summary statistics to consumers. We treat the platform as a passive information aggregator, abstracting from platform incentives.

3.2 Product Characteristics

Product quality Q_i is drawn independently from a normal distribution $Q_i \sim \mathcal{N}(\mu_Q, \sigma_Q^2)$. Quality is unobserved by consumers prior to purchase. The normal distribution has unbounded support, while our rating scale is $[1, 5]$. For reasonable parameter values, extreme realizations outside $[1, 5]$ have negligible probability. Alternatively, one may interpret Q as unbounded latent quality that maps to observed ratings through the bounded rating function. Quality captures the objective performance of the product, including durability, functionality, and design. We assume quality is fixed and identical for all units of product i.

Each product i has an observable raw price $p_i \in \mathbb{R}_+$. Consumers perceive price on a category-normalized scale $\tilde{p}_i = f_c(p_i) \in [1, 5]$, where $f_c : \mathbb{R}_+ \to [1, 5]$ is a category-specific, strictly increasing function. The normalization captures the intuition that price interpretation is context-dependent. A €300 smartphone occupies a different position in its category than a €300 automobile. The function f_c maps raw prices to expected quality on the five-point rating scale.

The mispricing of product i is $\eta_i \equiv \tilde{p}_i - Q_i$. When $\eta_i > 0$, the product is a lemon (overpriced relative to quality). When $\eta_i < 0$, the product is a bargain (underpriced relative to quality). When $\eta_i \approx 0$, the product is correctly priced. We assume mispricing is normally distributed with mean zero: $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$, independent of Q_i. Mispricing arises from various sources consistent with passive sellers under perfect competition: measurement error in quality assessment, category-specific normalization errors, temporary promotional pricing misaligned with true quality, or idiosyncratic pricing noise. We do not require strategic markup behavior to generate mispricing. These assumptions together imply that normalized price is a noisy signal of quality: $\tilde{p}_i = Q_i + \eta_i$. This is the key relationship that links observable prices to unobservable quality.

3.3 Basic Assumption

The entire model is based on the following key assumption: ratings do not measure absolute product quality, but rather the gap between expectations and realized quality. Consumers form expectations based on observable characteristics, and after consumption, they compare the experienced quality with these expectations. A product that is inexpensive (e.g., $Q = 2$ and performs as expected (or slightly better) may therefore receive a high rating (e.g., 5 stars), because it exceeds or meets a low reference point. In contrast, a more expensive product (e.g., $Q = 3$) sets higher expectations; if the realized quality falls short of these expectations, the consumer experiences disappointment and may assign a low rating (e.g., 2 stars), despite the product being objectively of higher quality. Consequently, ratings reflect subjective expectation gaps rather than objective quality levels, which implies that higher-quality or higher-priced products are not necessarily associated with higher average ratings.

3.4 Loop

Each period t proceeds in four stages. In Stage 1 (Information and beliefs), consumer j arrives and observes $(p_i, \bar{r}_{it}, n_{it})$ for all products, then forms expected quality r_{it}^* (the reference point) for each product. In Stage 2 (Demand and Experience), consumer j chooses which product to purchase (or chooses the outside option), and if she decides to buy it, experiences quality q_{ijt} and computes the gap $\Delta_{ijt} = q_{ijt} - r_{it}^*$. In Stage 3 (Posting and Rating), consumer decides whether to post a review; if posting, she assigns a star rating $r_{ijt} \in \{1, 2, 3, 4, 5\}$. In Stage 4 (Aggregation), the platform updates the public information $n_{i,t+1}, \bar{r}_{i,t+1}$ based on any new review.

4 Stage 1: Information and Reference Point Formation

Upon arrival, consumer j observes three things in the platform: the price of the product (p_i), the average rating of the product at time t ($\bar{r}_{it}$), and the number of reviews that the product has at time t (n_{it}). She uses this information to form an expectation of her experienced quality, a reference point, for each product.

4.1 Bayesian Foundation

It is important to note that the buyer has a completely non-informative prior, that is, without looking at any of the three pieces of data provided by the platform, the buyer cannot know anything about the quality of the product, or what they know has no precision at all ($\tau_0 \approx 0$). Consequently, the buyer will rely solely on the signal when assessing the expected quality of the product. Using Bayesian updating, and considering s as the signal, we have:

$$E[Q \mid s] = \frac{\tau_0}{\tau_0+\tau_s}\mu_0 + \frac{\tau_s}{\tau_0+\tau_s}s = s.$$

The posterior mean is going to be the signal itself. We now derive it.

4.2 Reference Point Formation

From all three elements that made the signal, only the price p_i provides direct information about the product's quality to the buyer. Specifically, and as presented in 3.2, the buyer observes p_i and forms $\tilde{p}_i$, which offers an expectation of the product's quality based on the price in a scale from 1 to 5. Additionally, and given that the average rating $\bar{r}$ is an indicator of the "satisfaction", it can be approximated as: $\bar{r} \approx 5 - \kappa\eta + \varepsilon$, where η reflects the mispricing, $\kappa > 0$ captures how strongly mispricing shifts average ratings, and $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2/n)$ represents the noise from finite samples and match heterogeneity. Inverting the rating-mispricing relationship, the rating gap $(\bar{r}-5)$ is a signal of $-\eta$: $\hat{\eta} = \frac{1}{\kappa}(5-\bar{r})$. Knowing that the price is not always reliable, due to mispricing, the price signal is $\tilde{p} = Q + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$, so the quality estimate is $\hat{Q} = \tilde{p} - \hat{\eta} = \tilde{p} + \frac{1}{\kappa}(\bar{r}-5) = \tilde{p} + \phi(\bar{r}-5)$, where $\phi = 1/\kappa$ is a positive constant. So the buyer derives the expected quality for product i by adjusting $\tilde{p}$ by subtracting the expected mispricing η, which is conditioned by the average rating. Formally, the posterior belief is:

$$r_{it}^* = E[Q_i \mid p_i, \bar{r}_{it}] = \tilde{p_i} - E[\eta_i \mid \bar{r}_{it}] = \tilde{p_i} + \phi(\bar{r}_{it} - 5)$$

The key idea is that, although ratings reveal mispricing rather than quality directly, price and ratings jointly reveal quality. A consumer can infer quality from the price and the average rating. This formula shows that if the average rating is 5 ($\bar{r} = 5$), then the buyer will fully rely on the price ($r^* = \tilde{p}$). However, if the average rating is lower than 5 ($\bar{r} < 5$), which would indicate overpricing, the buyer will lower their expectations about the quality of product ($r^* < \tilde{p}$). Notice that the signal, the expected quality, serves as a reference point for quality. Regarding the precision of the reference point estimate, τ_{r^*}, it is increasing in the number of reviews n, but in a diminishing manner. The precision will function similarly to this: $\tau_{r^*}(n) = \tau_{r^*}(n-1) + \lambda \ln(1+n)$, where $\tau_{r^*}(n-1)$ is the precision of the signal prior to the last review. $\tau_{r^*}(0) = 0$. All results require only that τ_{r^*} be increasing in n. The joint information $(\tilde{p}_i, \bar{r}_i)$ solves the Akerlof adverse selection problem. Lemons ($\eta > 0$) receive low $\bar{r}$, enabling buyers to infer $Q < \tilde{p}$. Bargains ($\eta < 0$) receive high $\bar{r}$, enabling buyers to infer $Q > \tilde{p}$. This is our first main result: ratings solve adverse selection not by revealing quality directly, but by revealing calibration, which combined with price identifies quality.

5 Stage 2: Demand and Experience

Given her reference points, the consumer chooses which product to purchase.

5.1 Demand Specification

The demand for product i at time t is $D_{it} = \lambda_0 + \gamma(\tau_{r^*_{it}}) \cdot \bar{r}_{it} - \delta \cdot p_i$, where $\lambda_0 > 0$ is baseline demand, $\gamma(\tau_{r^*_{it}})$ is sensitivity to average rating (increasing in precision $\tau_{r^*_{it}}$), and $\delta > 0$ is price sensitivity.The second term indicates that demand is higher for products with high average rating, and lower for those with low average ratings. Additionally, note that the weight $\gamma(\tau_{r^*})$ in which the average rating affects demand will depend on the precision of the signal. When few reviews exist, consumers discount rating information; when many reviews exist, ratings strongly influence demand. The demand structure implies that lemons lose market share endogenously. Products with $\eta > 0$ (lemons) experience declining demand over time, while products with $\eta \leq 0$ (correctly priced or bargains) maintain or increase demand.

5.2 Experience and the Gap

If the buyer decides to purchase the product, she will experience it. Consumer j's experience with product i at time t is $q_{ijt} = Q_i + m_{ijt}$, where $m_{ijt} \sim \mathcal{N}(0, \sigma_m^2)$ is an idiosyncratic match shock, independent across consumers, products, and time. The match shock captures individual variation in product-consumer fit: taste differences, usage context, unit-to-unit variation in manufacturing, delivery conditions, or random factors affecting the consumption experience. Even a high-quality product will occasionally disappoint a particular consumer; even a low-quality product will occasionally delight. The experience-expectation gap for consumer j purchasing product i at time t is $\Delta_{ijt} \equiv q_{ijt} - r^*_{it}$. The gap measures how much the consumer's experience deviated from her expected experienced quality (the reference point). Positive gaps indicate pleasant surprises; negative gaps indicate disappointments.

6 Stage 3: Posting and Rating

This is the core of our model. When a consumer j has purchased the product and experienced its quality q_{ijt}, she can decide whether or not to post a rating, and if so, which rating to give it. In the model we propose, we build on the idea that both decisions are reference-dependent; that is, both the decision of whether to post your opinion and the decision of which rating to assign, in the case of posting, depend on the gap $\Delta_{ijt} = q_{ijt} - r^*_{it}$.

6.1 Posting decision

The utility derived from the expectation gap is based on the Prospect Theory Utility. So if a buyer has a gap Δ, their utility is going to be:

$$u(\Delta) = \begin{cases} \beta_+ \Delta^\alpha & \Delta \geq 0 \\ -\beta_- |\Delta|^\alpha & \Delta < 0 \end{cases}$$

With $\beta_- > \beta_+$ (loss aversion) and $\alpha \in (0, 1)$ (diminishing sensitivity). This formula captures the fact that the larger the gap, the greater the utility. Moreover, if the gap is negative, this generates more utility (in absolute value) than a positive gap of the same size, so the impact of a loss is felt more intensely than an equivalent gain. In this case, utility refers to the willingness of a buyer to post their opinion about a given product. Based on this, we assume that a buyer j will post if and only if:

$$|u(\Delta_{ijt})| \geq d_j$$

That is, if the utility of the gap exceeds their constant d_j. Each consumer j has a posting cost d_j drawn from $d_j \sim \mathcal{N}(\mu_d, \sigma_d^2)$, independent of all other random variables. It captures all individual's predisposition to post, as well as the cost of posting, for example in terms of time or effort. The probability that a consumer with gap Δ posts a review is therefore $\pi(\Delta) = \Phi\left(\frac{|u(\Delta)| - \mu_d}{\sigma_d}\right)$, where Φ is the standard normal CDF.

6.2 Rating

If the condition of posting is met and the buyer decides to post, the rating will depend only on the gap between experience and expectation (Δ), as follows:

$$R(\Delta) = \begin{cases} 5, & \Delta \geq 0 \\ 4, & -\frac{c}{3} \leq \Delta < 0 \\ 3, & -\frac{2c}{3} \leq \Delta < -\frac{c}{3} \\ 2, & -c \leq \Delta < -\frac{2c}{3} \\ 1, & \Delta < -c \end{cases}$$

Where $c > 0$ is the disappointment threshold. So, if the gap is zero or positive, the buyer will give 5 stars. If the gap is negative, the rating will depend on how negative the gap is, resulting in 4, 3, 2, or 1 star accordingly. Finally, the last important concept is the probability of giving a rating r, where $r \in \{1, 2, 3, 4, 5\}$. This probability depends on both the probability of posting given the gap (Δ) and the distribution of delta itself $f(\Delta)$:

$$\mathbb{P}(R = r) = \frac{\int_{\text{Bin}_r} \pi(\Delta) \cdot f(\Delta)\, d\Delta}{\int_{-\infty}^{\infty} \pi(\Delta) \cdot f(\Delta)\, d\Delta}$$

The derivation of this formula is provided in Appendix A.

7 Stage 4: Aggregation and Feedback

After each period, the platform updates the public state based on any new review. If consumer j posts review r_{ijt} for product i at time t, then $n_{i,t+1} = n_{it} + 1$ and $\bar{r}_{i,t+1} = \frac{n_{it} \cdot \bar{r}_{it} + r_{ijt}}{n_{it}+1}$. If no review is posted, $n_{i,t+1}, \bar{r}_{i,t+1} = n_{it}, \bar{r}_{it}$.

The four stages create a feedback loop. Reviews affect future expectations (reference points), which affect demand, which determines who experiences the product and posts future reviews. This feedback loop drives the dynamic evolution of the J-curve.

8 Static J-Curve

In sections 4 through 7, we have presented the feedback loop that explains how expectations are formed, certain products are bought, a gap between the expected quality and the quality experienced is generated, how the decision to post a review comes about, and how reviews are aggregated by the platform. Now, we turn our attention to the very first iteration of this loop, when there is no previous reviews for product i, that is, when $n = 0$.

In this scenario, given that there are no previous reviews, the reference point (r^*_{it}) used to generate an expectation of the quality of product i is only made of $\tilde{p}_i$. Recall:

$$r^*_{it} = E[Q_i \mid p_i, \bar{r}_{it}] = \tilde{p}_i - E[\eta_i \mid \bar{r}_{it}] = \tilde{p}_i + \underbrace{\phi(\bar{r}_{it} - 5)}_{\text{This term is non-existent}} = \tilde{p}_i = Q_i + \eta_i$$

Therefore $\Delta_0 = q - r^* = (Q + m) - (Q + \eta) = m - \eta$. Since $m \sim \mathcal{N}(0, \sigma^2_m)$ and $\eta \sim \mathcal{N}(0, \sigma^2_\eta)$ are independent, we have $\Delta_0 \sim \mathcal{N}(0, \sigma^2_m + \sigma^2_\eta)$. Quality Q cancels, leaving only match noise and mispricing. Figure 1 represents the distribution of Δ in this setting.

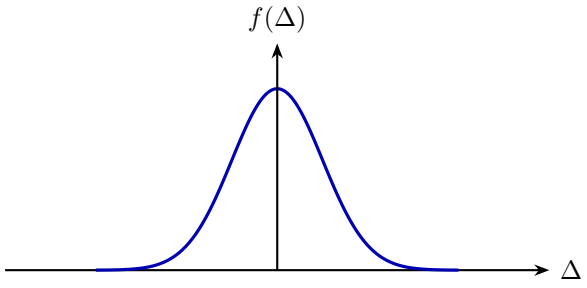


Figure 1: Distribution of Δ for the static scenario.

The posting probability introduced in section 6 has a distinct shape. At $\Delta = 0$, we have $\pi(0) = \Phi(-\mu_d/\sigma_d) < 0.5$ for $\mu_d > 0$ (low probability at zero gap). As $|\Delta| \to \infty$, we have $\pi(\Delta) \to 1$ (high probability in tails). Under loss aversion ($\beta_- > \beta_+$), we have $\pi(-\delta) > \pi(\delta)$ for all $\delta > 0$. Hence $\pi(\Delta)$ is not exactly U-shaped in Δ, given that the left branch is slightly elevated relative to the right branch.

In Figure 2 a graphical approximation of how $\pi(\Delta)$ in Δ looks like is provided. It is important to note that the magnitude by which the left tail is larger than the right one is exaggerated in this plot for explainability purposes. In reality, the effects of loss aversion are likely more subtle.

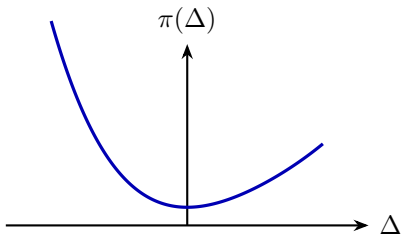


Figure 2: Probability of posting conditional on Δ for the static scenario.

As explained in section 6, the distribution of Δ and the probability of posting given Δ together make $\mathbb{P}(R = r)$, that is, the probability for each rating bin. Recall:

$$\mathbb{P}(R = r) = \frac{\int_{\text{Bin}_r} \pi(\Delta) \cdot f(\Delta)\, d\Delta}{\int_{-\infty}^{\infty} \pi(\Delta) \cdot f(\Delta)\, d\Delta}$$

The key asymmetry is that all non-negative gaps (Δ) share a single bin (5 stars), while negative gaps spread across four bins. The bin $[0, \infty)$ has infinite width, while the four negative bins partition $(-\infty, 0)$. This asymmetric mapping from symmetric gaps to ratings is the mechanical source of the J-curve.

Additionally, and under loss aversion, $\pi(-\delta) > \pi(\delta)$ for $\delta > 0$. This amplifies extreme negatives relative to extreme positives. For bin 1 (extreme negatives, $\Delta < -c$) versus bin 2 (mild negatives, $\Delta \in [-c, -2c/3)$), bin 1 contains gaps with large $|\Delta| > c$, which have high $\pi(\Delta)$, while bin 2 contains gaps with moderate $|\Delta| \in [2c/3, c]$, which have lower $\pi(\Delta)$. Defining $\tilde{M}_r = \int_{\text{Bin}_r} \pi(\Delta) f(\Delta) d\Delta$ as the posting-weighted mass in bin r; for $c/\sigma_\Delta \approx 1$, the tail mass $M_1 \approx 0.16$ while $M_2 \approx 0.09$. Loss aversion further amplifies bin 1 relative to bin 2 because extreme disappointments generate higher posting utility than moderate disappointments. Thus $\tilde{M}_1 > \tilde{M}_2$, and after normalization, $\mathbb{P}(R = 1) > \mathbb{P}(R = 2)$.

This is a key result. The J-curve does not arise from asymmetric gaps; it arises from asymmetric rating bins. The J-curve exists from the first review ($n = 1$). No learning or dynamic accumulation is required.

9 Dynamic J-Curve

We now analyze the long-run evolution of the rating distribution, that is, when $n \to \infty$.

9.1 Product Sorting

In the long run, products sort into two groups based on mispricing: survivors (correctly priced products and bargains), and lemons (overpriced items). Correctly priced products and bargains ($\eta \leq 0$) accumulate reviews. Their gap distribution converges to $\Delta \sim \mathcal{N}(\mu_+, \sigma_m^2)$ with $\mu_+ \geq 0$. Lemons ($\eta > 0$) stagnate at low review counts. Their gap distribution remains centered below zero. For correctly priced products ($\eta = 0$), as $n \to \infty$, $\bar{r}$ converges to its population mean. Since

$\mathbb{E}[\Delta] = 0$, we have $\bar{r} \to 5$. The reference point $r^* = \tilde{p} + \phi(\bar{r} - 5) \to \tilde{p} = Q$. Gaps become $\Delta = Q + m - Q = m \sim \mathcal{N}(0, \sigma_m^2)$, centered at zero.

For bargains ($\eta < 0$), even as $n \to \infty$, the average rating cannot exceed 5 (the maximum of the scale). Hence $\bar{r}$ converges to 5, and $r^* \to \tilde{p} = Q + \eta < Q$. The gap is $\Delta = q - r^* = (Q+m) - (Q+\eta) = m - \eta$, with mean $-\eta > 0$. Bargains generate persistently positive mean gaps ($\mu_+ = -\eta > 0$), explaining the flood of 5-star reviews for underpriced goods.

For lemons ($\eta > 0$), bad reviews ($\bar{r} < 5$) reduce demand through $\gamma(\tau) \cdot \bar{r}$. Lower demand means fewer purchases and slower review accumulation. The product stagnates with persistently low ratings.

9.2 J-Curve Sharpening

The J-curve sharpens over time. As the number of reviews $n \to \infty$, the distribution of gaps $f(\Delta)$ shifts: the mass for $\Delta \geq 0$ increases, while the mass for $\Delta < 0$ decreases. Consequently, the probability of observing a 5-star rating, $\mathbb{P}(R = 5)$, rises as the survivors—correctly priced products and bargains—dominate the review volume. Conversely, the probability of a 1-star rating, $\mathbb{P}(R = 1)$, declines slightly but remains positive due to match noise σ_m^2. Intermediate ratings (2, 3, and 4 stars) become increasingly rare, producing a sharper J-shaped rating distribution over time.

As survivors accumulate reviews and lemons stagnate, the review-weighted gap distribution shifts rightward. More mass falls in bin 5, increasing $\mathbb{P}(R = 5)$. Match noise $m \sim \mathcal{N}(0, \sigma_m^2)$ ensures that even survivors occasionally have $\Delta = m - \eta < -c$ (for correctly priced products with $\eta = 0$, this requires $m < -c$). Such realizations generate 1-star reviews. The 1-star tail persists but shrinks as survivors dominate.

10 Empirical Predictions

Our model generates three testable predictions.

P1: J-Curve at $n = 1$. In the static model, the distribution of first reviews is J-shaped. This can be tested by examining rating distributions conditional on $n = 1$.

P2: J-Curve Sharpens with n. The probability $\mathbb{P}(R = 5 \mid n)$ is increasing in review count n. This can be tested by examining rating distributions as $n \to \infty$.

P3: Lemons Stagnate. Products that are thought to be overpriced have lower review counts in the long-term. This can be tested by computing the share of low-rated products (lemons) as the number of reviews (n) increases.

11 Empirical Analysis

In order to test our empirical predictions, we use a dataset of Amazon product reviews from May 1996 to September 2023, which contains about 570 million reviews (Hou et al., 2024).

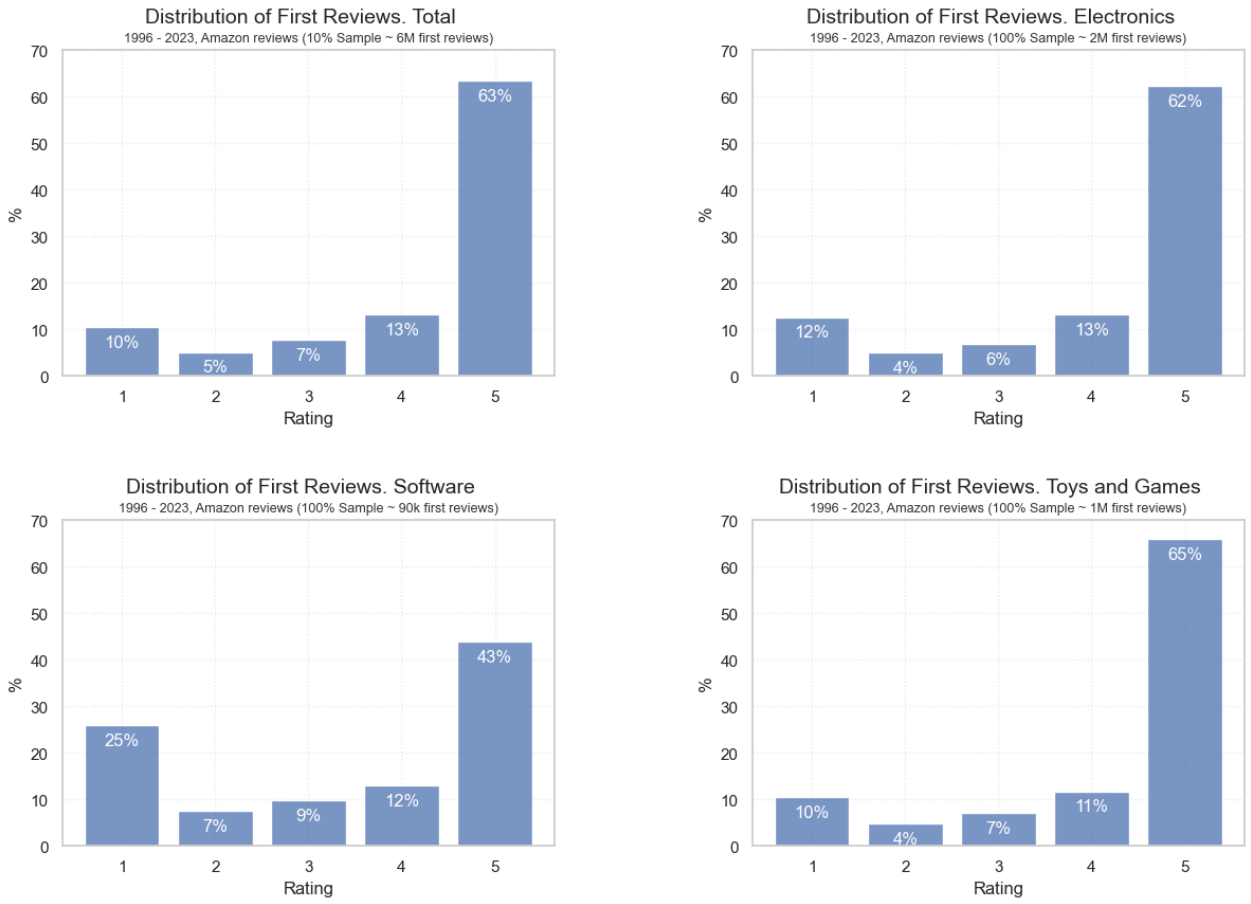


Figure 3: J-curve distributions for first reviews across product categories.

P1: J-Curve at $n = 1$. The first prediction introduced in the previous section is confirmed by Figure 3, where the J-Curve at $n = 1$ is displayed. In the upper-left panel, the distribution of the first reviews is displayed for a representative 10-percent sample of the reviews across categories. Note that the sampling was introduced to make the J-Curve graphically visible. The upper-right, lower-left, and lower right plots display the distribution of first reviews for specific categories, namely: electronics, software, and toys and games, respectively. These categories were selected to illustrate distinct cases while maintaining validity through a large volume of reviews. Overall, and although the software category seems to exhibits a less pronounced J-Curve, all four cases presented above show the presence of the J-curve from the very first review.

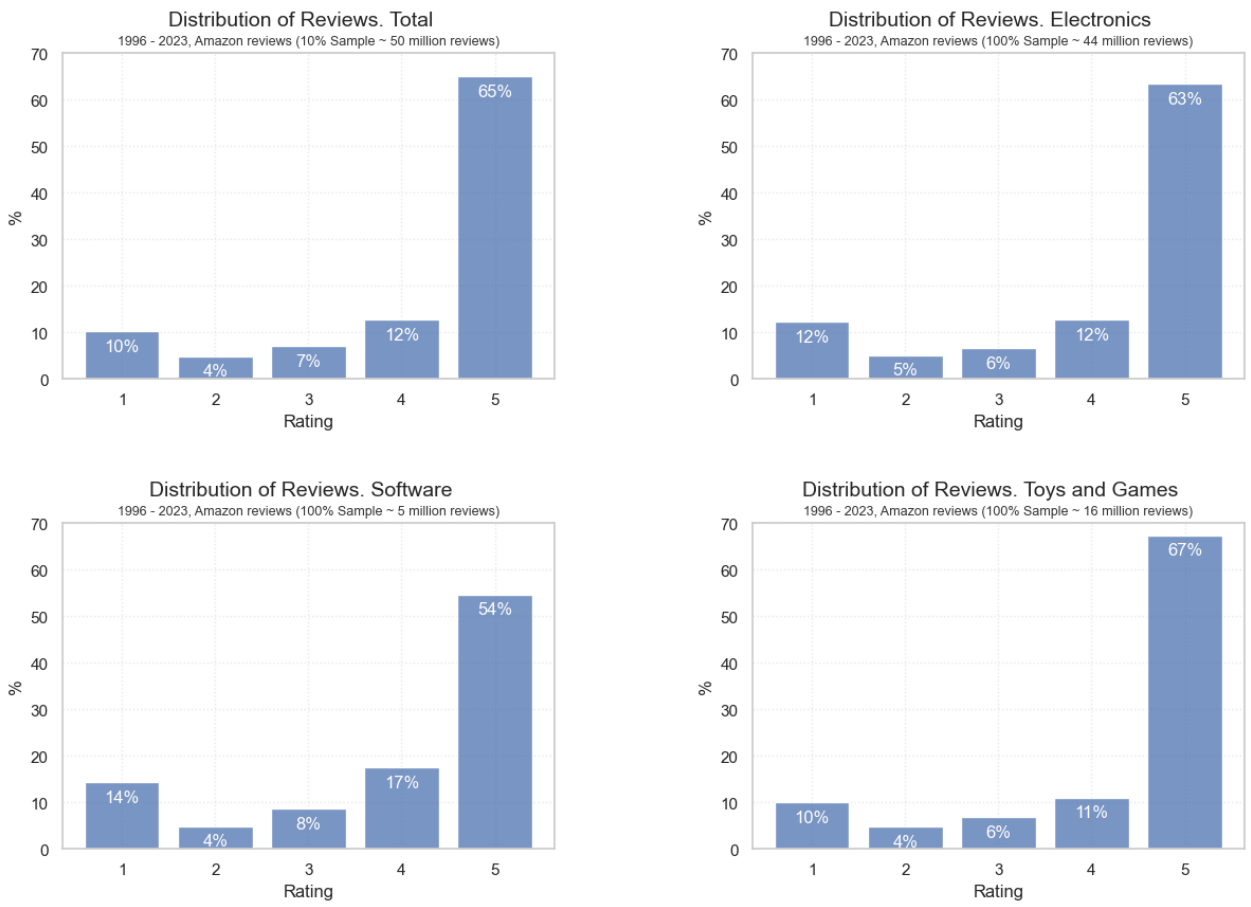


Figure 4: J-curve sharpening with review count.

P2: J-Curve Sharpens with n. Similarly to the previous scenario, in Figure 4 we show the distribution of reviews without any constraint on the number of reviews for the same four cases. As we predicted, the J-Curve appears to sharpen over the number of reviews, although the observed effects are milder than expected. The share of 5-star ratings increases about 1-2 percentage points for all cases except software, which experiences a notable increase of 11 percentage points. The rest of the rating bins remain largely unchanged across categories, although in software we observe adjustments in the share of rating bins 1 and 4, which ultimately bring the distribution closer to a J-Curve, and closer to the rest of the cases evaluated.

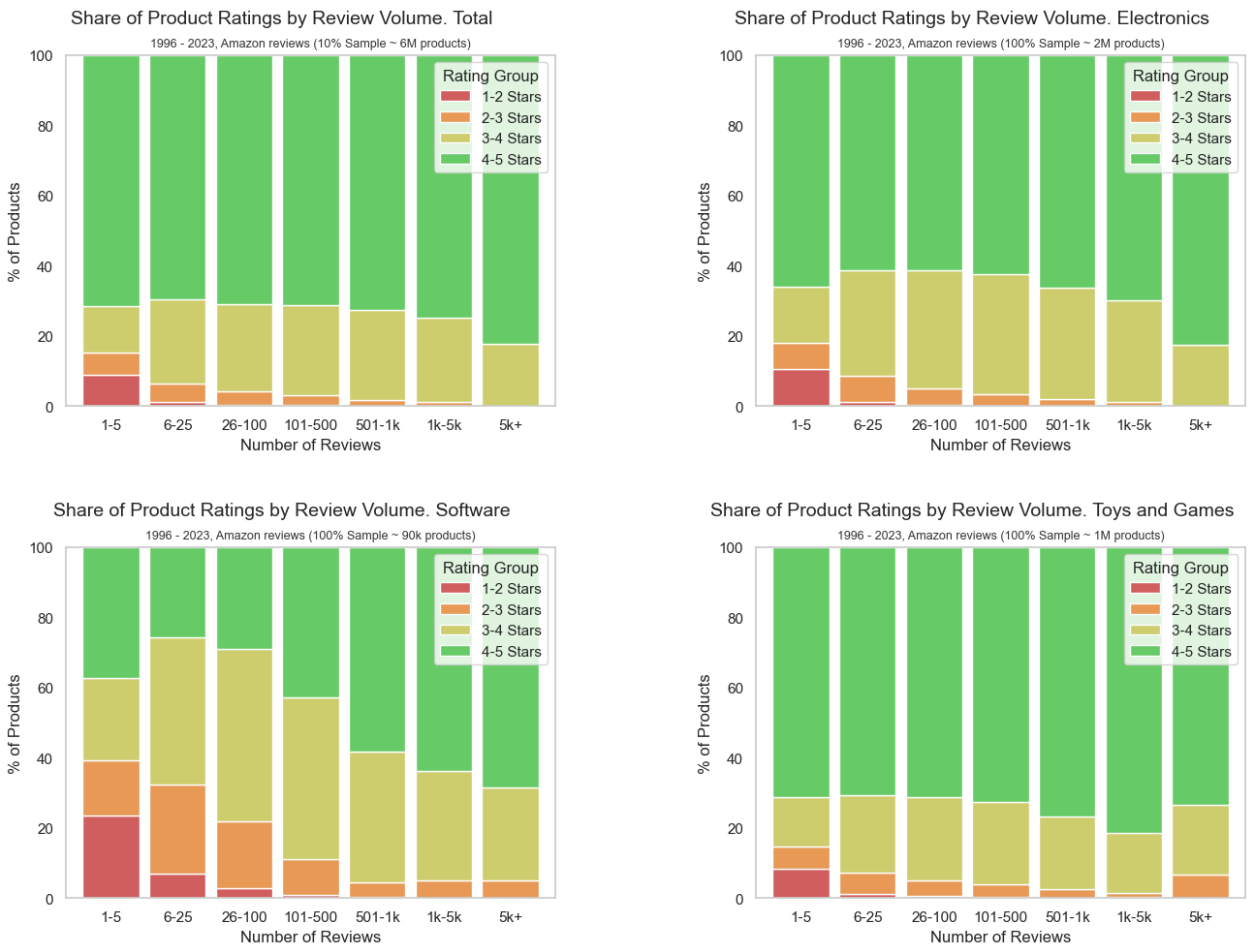


Figure 5: Share of product ratings by review volume.

P3: Lemons Stagnate. Figure 5 presents four plots showing the share of products per rating bin as the total number of reviews increases. As can be seen in all four cases, the share of the two lowest rating groups (associated with more extreme lemons (1–2 stars and 2–3 stars)) declines as review volume grows. This confirms the third empirical prediction introduced in the previous section: in the long term, ratings for lemons stagnate as demand for these products diminishes.

12 Conclusion

This paper has analyzed the emergence and persistence of the J-shaped distribution in online ratings. We have addressed two related questions: first, why do online ratings exhibit a J-curve rather than a normal distribution, and second, why does this pattern persist across products, platforms, and time? We have provided the following answers. First, the J-curve emerges because ratings measure calibration rather than quality: consumers rate the gap between experienced and expected quality, and the mapping from continuous gaps to discrete stars is asymmetric. All positive gaps are assigned to a single bin, while negative gaps are distributed across four bins. Second, the J-curve persists because market selection reinforces it: correctly priced products accumulate reviews as

favorable ratings attract demand, while overpriced products stagnate, concentrating review mass at the extremes over time. These answers are linked by a common framework in which consumers form expectations from prices and prior ratings, post reviews when experience deviates substantially from expectations, and assign ratings that reflect the sign and magnitude of the gap filtered through asymmetric binning.

We have substantiated this framework by means of a reference-dependent utility model and an empirical analysis of more than 500 million Amazon reviews spanning three decades. The model generates closed-form predictions for the shape of the rating distribution at market entry and in the long run, for the relationship between prices and average ratings, and for the dynamics of review accumulation across product types. The empirical analysis confirms the presence of the J-curve across product categories and time periods, documents its sharpening as products age, and establishes the predicted stagnation of poorly rated products.

The economics of online ratings deserves further investigation. Theoretically, natural extensions include endogenizing seller pricing in response to ratings, modeling platform design of rating technologies, and analyzing strategic manipulation of reviews. Our empirical analysis has focused on rating distributions and review dynamics, leaving direct tests of the calibration mechanism to future work. The data contain prices, which would permit tests of the joint price-rating predictions central to the model. Experimental evidence, in which expectations are manipulated exogenously, would provide further identification.

The questions addressed in this paper, how decentralized information aggregation shapes market outcomes, how asymmetric institutions distort signals, how selection reinforces initial patterns, are not confined to e-commerce. They arise wherever agents form expectations, experience outcomes, and communicate evaluations: labor markets, financial markets, political accountability. The J-curve is, in this light, a particular instance of a more general phenomenon.

References

Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3):488–500.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2012). Salience theory of choice under risk. *Quarterly Journal of Economics*, 127(3):1243–1285.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2013). Salience and consumer choice. *Journal of Political Economy*, 121(5):803–843.

Hou, Y., Li, J., He, Z., Yan, A., Chen, X., and McAuley, J. (2024). Bridging Language and Items for Retrieval and Recommendation. *arXiv preprint arXiv:2403.03952*.

Hu, N., Pavlou, P. A., and Zhang, J. (2009). Overcoming the J-shaped distribution of product reviews. *Communications of the ACM*, 52(10):144–147.

Kahneman, D. and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291.

Kőszegi, B. and Rabin, M. (2006). A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4):1133–1165.

A Derivation of the Probability of Rating r

Let $R \in \{1, 2, 3, 4, 5\}$ denote the rating given by a consumer. The probability that a consumer posts a rating r depends on both the probability of posting given the gap, $\pi(\Delta)$, and on the distribution of gaps, $f(\Delta)$.

If a consumer has gap Δ and decides to post a review, then the rating is entirely determined by which bin Δ falls into:

$$\Pr(R = r \mid \Delta, \text{posted}) = \mathbf{1}_{\{\Delta \in \text{Bin}_r\}},$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function.

Without conditioning on whether the consumer posts, the probability of giving rating r given Δ is:

$$\Pr(R = r \mid \Delta) = \pi(\Delta) \cdot \mathbf{1}_{\{\Delta \in \text{Bin}_r\}}.$$

Integrating over the distribution of gaps gives the unnormalized probability of rating r:

$$\int_{-\infty}^{\infty} \pi(\Delta)\, \mathbf{1}_{\{\Delta \in \text{Bin}_r\}}\, f(\Delta)\, d\Delta = \int_{\text{Bin}_r} \pi(\Delta)\, f(\Delta)\, d\Delta.$$

Finally, dividing by the total probability that a consumer posts a review gives the conditional probability of observing rating r:

$$\Pr(R = r) = \frac{\int_{\text{Bin}_r} \pi(\Delta)\, f(\Delta)\, d\Delta}{\int_{-\infty}^{\infty} \pi(\Delta)\, f(\Delta)\, d\Delta}. \tag{1}$$

This derivation clarifies that the observed rating distribution depends both on the likelihood of posting given the gap and on the distribution of gaps across consumers.

B Counterfactual Analysis

Our model combines several assumptions to generate the J-curve. This appendix systematically "shuts down" each assumption to identify its marginal contribution. We derive analytical results for four counterfactuals.

B.1 Baseline Recap

The baseline model produces the J-curve through four mechanisms. First, asymmetric bins: $R(\Delta) = 5$ for all $\Delta \geq 0$; four bins for $\Delta < 0$. Second, loss aversion: $\beta_- > \beta_+$ in the posting utility. Third, selective posting: post if $|u(\Delta)| \geq d_j$, filtering moderate experiences. Fourth, increasing precision: $\tau_{r^*} \propto \ln(1 + n)$ affects demand weight $\gamma(\tau_{r^*})$.

B.2 Counterfactual 1: Symmetric Rating Bins

We replace asymmetric bins with symmetric bins of equal structure on both sides of zero. We keep the same gap distribution, posting rule, and loss aversion; only the mapping $\Delta \mapsto R$ is symmetrized. The symmetric rating bins are: $R^{\text{sym}}(\Delta) = 5$ if $\Delta > c$, $R^{\text{sym}}(\Delta) = 4$ if $\Delta \in (c/3, c]$, $R^{\text{sym}}(\Delta) = 3$ if $\Delta \in (-c/3, c/3]$, $R^{\text{sym}}(\Delta) = 2$ if $\Delta \in (-c, -c/3]$, and $R^{\text{sym}}(\Delta) = 1$ if $\Delta \leq -c$.

Under symmetric bins, bins 5 and 1 are symmetric tails ((c, ∞) and $(-\infty, -c]$), bins 4 and 2 are symmetric intermediate ($(c/3, c]$ and $(-c, -c/3]$), and bin 3 is centered ($(-c/3, c/3]$).

With symmetric rating bins and symmetric gap distribution $f(\Delta) = f(-\Delta)$, if $\pi(\Delta) = \pi(-\Delta)$ (no loss aversion), the rating distribution is symmetric around $R = 3$. If $\beta_- > \beta_+$ (loss aversion), the rating distribution is left-skewed (mode at 1, not 5). In neither case is the distribution right-skewed J-shaped (mode at 5).

The proof proceeds as follows. Define unweighted bin masses $M_r = \int_{\text{Bin}_r} f(\Delta)d\Delta$. By symmetry of f and the bin structure, $M_5 = \int_c^{\infty} f(\Delta)d\Delta = \int_{-\infty}^{-c} f(\Delta)d\Delta = M_1$ and $M_4 = \int_{c/3}^{c} f(\Delta)d\Delta = \int_{-c}^{-c/3} f(\Delta)d\Delta = M_2$.

In the no loss aversion case, with $\pi(\Delta) = \pi(-\Delta)$, we have $\tilde{M}_5 = \int_c^{\infty} \pi(\Delta)f(\Delta)d\Delta = \int_{-\infty}^{-c} \pi(-\Delta)f(-\Delta)d\Delta = \tilde{M}_1$. Similarly $\tilde{M}_4 = \tilde{M}_2$. The distribution is symmetric around $R = 3$.

With loss aversion, $\pi(-\delta) > \pi(\delta)$ for $\delta > 0$. For $\Delta^* > c$ (in bin 5) and $-\Delta^* < -c$ (in bin 1), we have $\pi(-\Delta^*) > \pi(\Delta^*)$. So negative gaps receive higher posting weight: $\tilde{M}_1 = \int_{-\infty}^{-c} \pi(\Delta)f(\Delta)d\Delta > \int_c^{\infty} \pi(\Delta)f(\Delta)d\Delta = \tilde{M}_5$. The distribution is left-skewed with $\mathbb{P}(R = 1) > \mathbb{P}(R = 5)$, the opposite of the J-curve.

Asymmetric bins are necessary. Symmetric bins with symmetric f yield symmetric or left-skewed distributions, never a right-skewed J-shape with mode at 5.

B.3 Counterfactual 2: No Loss Aversion

We set $\beta_+ = \beta_- = \beta$, so posting utility satisfies $|u(\Delta)| = \beta|\Delta|^{\alpha}$ for all Δ. We keep asymmetric bins, selective posting, and precision dynamics; only loss aversion is removed.

With asymmetric rating bins and $\beta_+ = \beta_- = \beta$, the J-curve persists ($\mathbb{P}(R = 5) > \mathbb{P}(R = r)$ for $r \in \{1, 2, 3, 4\}$), the 1-star tail is weaker ($\mathbb{P}^{\text{no LA}}(R = 1) < \mathbb{P}^{\text{baseline}}(R = 1)$), and the ratio $\mathbb{P}(R = 1)/\mathbb{P}(R = 5)$ decreases relative to baseline.

The J-curve arises from bin asymmetry. With asymmetric bins, $M_5 = \int_0^{\infty} f(\Delta)d\Delta = 0.5$ while $M_1 + M_2 + M_3 + M_4 = 0.5$. Even with symmetric π (no LA), bin 5 captures all $\Delta \geq 0$, ensuring $\tilde{M}_5 > \tilde{M}_r$ for each $r \in \{1, 2, 3, 4\}$.

Under baseline (with LA), for any $\delta > 0$, $|u^{\text{LA}}(-\delta)| = \beta_-\delta^{\alpha} > \beta_+\delta^{\alpha} = |u^{\text{LA}}(\delta)|$, so $\pi^{\text{LA}}(-\delta) > \pi^{\text{LA}}(\delta)$: negative gaps have higher posting probability. Without LA (setting $\beta_+ = \beta_- = \beta$), $|u^{\text{no LA}}(-\delta)| = \beta\delta^{\alpha} = |u^{\text{no LA}}(\delta)|$, so $\pi^{\text{no LA}}(-\delta) = \pi^{\text{no LA}}(\delta)$: equal posting probability.

For bin 1 (extreme negatives, $\Delta < -c$), $\tilde{M}_1^{\text{LA}} = \int_{-\infty}^{-c} \pi^{\text{LA}}(\Delta)f(\Delta)d\Delta > \int_{-\infty}^{-c} \pi^{\text{no LA}}(\Delta)f(\Delta)d\Delta = \tilde{M}_1^{\text{no LA}}$. (The inequality holds because $\pi^{\text{LA}}(\Delta) > \pi^{\text{no LA}}(\Delta)$ for $\Delta < 0$.) After normalization, $\mathbb{P}^{\text{no LA}}(R = 1) < \mathbb{P}^{\text{LA}}(R = 1)$.

For bin 5 (all $\Delta \geq 0$), LA reduces posting probability (since $\beta_+ < \beta_-$ means positive gaps trigger

posting less). Without LA, bin 5 gains relative share. Combined with bin 1 losing share, the ratio $\mathbb{P}(R=1)/\mathbb{P}(R=5)$ decreases.

Loss aversion amplifies the 1-star tail but is not necessary for the J-curve itself. The J-curve exists without LA; only the tail prominence changes.

B.4 Counterfactual 3: Universal Posting

We set $\pi(\Delta) = 1$ for all Δ (everyone posts regardless of gap magnitude). We keep asymmetric bins and loss aversion parameters; only the selective posting mechanism is removed.

With asymmetric rating bins and $\pi(\Delta) = 1$, the J-curve persists ($\mathbb{P}(R=5) = 0.5 > \mathbb{P}(R=r)$ for $r \neq 5$), the middle fills in ($\mathbb{P}^{\text{univ}}(R \in \{2,3,4\}) > \mathbb{P}^{\text{baseline}}(R \in \{2,3,4\})$), and the tails weaken ($\mathbb{P}^{\text{univ}}(R=1) < \mathbb{P}^{\text{baseline}}(R=1)$).

With $\pi(\Delta) = 1$, $\mathbb{P}(R=r) = M_r = \int_{\text{Bin}_r} f(\Delta)d\Delta$. For $f(\Delta) = \mathcal{N}(0, \sigma_\Delta^2)$, $\mathbb{P}(R=5) = M_5 = \int_0^\infty f(\Delta)d\Delta = 0.5$. Since $\sum_{r=1}^4 M_r = 0.5$ and this is split among 4 bins, each $M_r < 0.5$. The J-curve exists from bin asymmetry alone.

Under baseline selective posting, $\pi(\Delta)$ is U-shaped: low near $\Delta = 0$, high in tails. This means moderate $|\Delta|$ (bins 2, 3, 4) get down-weighted by low $\pi(\Delta)$, while extreme $|\Delta|$ (bins 1 and 5) get up-weighted by high $\pi(\Delta)$. Under universal posting, each bin receives exactly its raw mass M_r. The middle bins (2, 3, 4) recover their "fair share" of the gap distribution, filling in the hollow middle. Bin 1 (extreme negative tail) has high π under baseline due to the U-shape and loss aversion amplification. Under universal posting, bin 1 receives only its raw share $M_1 = \Phi(-c/\sigma_\Delta)$. Since baseline amplifies extremes, $\mathbb{P}^{\text{univ}}(R=1) = M_1 < \mathbb{P}^{\text{baseline}}(R=1)$.

Selective posting hollows the middle and amplifies both tails. Without it, the J-curve still exists but is less pronounced.

B.5 Counterfactual 4: Constant Demand Sensitivity

We set demand sensitivity $\gamma(\tau_{r^*}) = \gamma_0$ constant, removing the feedback mechanism where high-n products have ratings that more strongly influence demand. We keep asymmetric bins, loss aversion, and selective posting; only the precision dynamics are removed.

With constant $\gamma(\tau_{r^*}) = \gamma_0$, the static J-curve is unaffected (the distribution at $n = 0$ is identical to baseline), the J-curve does not sharpen with n ($\mathbb{P}(R = 5 \mid n)$ grows more slowly), and lemon weeding is slower (overpriced products stagnate less aggressively).

The static J-curve depends only on the gap distribution at $n = 0$ ($\Delta_0 = m - \eta \sim \mathcal{N}(0, \sigma_m^2 + \sigma_\eta^2)$), the posting function $\pi(\Delta)$, and the rating bins $R(\Delta)$. None of these depend on the demand sensitivity $\gamma(\tau_{r^*})$. The static J-curve is unchanged.

Dynamic sharpening occurs because survivors accumulate reviews faster than lemons. The mechanism is as follows: survivors have $\bar{r} \approx 5$, lemons have $\bar{r} < 5$; higher $\bar{r}$ increases demand through $\gamma(\tau_{r^*}) \cdot \bar{r}$; higher demand leads to more purchases and more reviews; more reviews lead to higher precision τ_{r^*} and higher $\gamma(\tau_{r^*})$.

Under baseline, the last step creates a compounding effect: products with more reviews have ratings

that matter more for demand. With constant γ, the last step is eliminated. The feedback from reviews to demand is linear rather than convex. Survivors still accumulate reviews faster, but the advantage does not compound. Sharpening is slower.

Lemon weeding requires bad ratings to reduce demand. Under baseline, $D_{\text{lemon}} = \lambda_0 + \gamma(\tau_{r^*}) \cdot \bar{r}_{\text{low}} - \delta p$. As n increases, $\gamma(\tau_{r^*})$ increases, amplifying the penalty from low $\bar{r}$. With constant γ_0, $D_{\text{lemon}} = \lambda_0 + \gamma_0 \cdot \bar{r}_{\text{low}} - \delta p$. The penalty is fixed. Lemons face less severe demand reduction and stagnate more slowly.

The learning mechanism (increasing $\gamma(\tau_{r^*})$) drives dynamic sharpening and lemon weeding but does not affect the static J-curve. The static J-curve exists at $n = 1$ regardless of precision dynamics.

B.6 Summary Table

The following table summarizes each assumption's contribution:

Scenario	Right-Skewed J (Mode at 5)?	Strong 1-Star?	Hollow Middle?	Dynamic Sharpening?
Baseline	Yes	Yes	Yes	Yes
(1) Symmetric bins	No	n/a	n/a	n/a
(2) No loss aversion	Yes	weaker	Yes	Yes
(3) Universal posting	Yes	weaker	No	Yes
(4) Constant γ	Yes	Yes	Yes	No

Key findings: Asymmetric bins are necessary and sufficient for the J-curve. Without them, no J-curve emerges regardless of other assumptions. Loss aversion amplifies the 1-star tail but is not necessary for the J-curve itself. Selective posting hollows the middle and amplifies both tails. The U-shape of $\pi(\Delta)$ is responsible for the characteristic hollow middle. Increasing demand sensitivity drives dynamics but does not affect the static J-curve.

C Extension

C.1 Winner-Take-All Dynamics

Our baseline model uses demand linear in ratings. An extension with attention convexity generates winner-take-all dynamics.

Under attention-weighted demand, $D_{it} \propto (1 + n_{it})^{\theta} \cdot \bar{r}_{it}^{\beta}$ with $\theta > 1$. Under this specification, small differences in early reviews are amplified over time. Products with lucky early draws attract disproportionate demand. Two products with identical quality can experience divergent trajectories.

The mechanism operates as follows. Consider two products with identical quality Q and identical prices, hence identical mispricing $\eta = 0$. At $t = 0$, both have $n = 0$ and $\bar{r} = 5$ by convention. Their expected gaps are both zero: $\mathbb{E}[\Delta] = 0$.

However, actual first reviews are stochastic. Product A might receive a positive match shock ($m_A > 0$) and hence a 5-star first review. Product B might receive a negative match shock ($m_B < 0$) and hence a 4-star or lower first review. After one period, product A has $\bar{r}_A = 5$ while product B has $\bar{r}_B < 5$.

Under attention-weighted demand, this small difference is amplified. Product A receives more demand in period 2, hence more reviews, hence higher precision, hence even more demand in period 3. Meanwhile, product B's lower rating reduces demand, slowing review accumulation and precision growth.

The divergence can be quantified. Let $n_A(t)$ and $n_B(t)$ denote review counts at time t. Under the attention-weighted demand specification, the ratio $n_A(t)/n_B(t)$ grows exponentially in t when $\theta > 1$. In the limit, product A dominates the market while product B becomes a "long tail" product with minimal visibility.

This winner-take-all dynamic explains several empirical patterns: the extreme concentration of reviews among a small number of products, the persistence of early rating advantages, and the difficulty new entrants face in competing with established products even when quality is comparable.

However, winner-take-all dynamics explain market concentration but not the J-curve. The J-curve exists without attention convexity; it is a property of the rating function, not demand feedback. The two phenomena are orthogonal: the J-curve describes the shape of the rating distribution, while winner-take-all describes the distribution of market shares.

C.2 Comparative Statics

We examine how the J-curve shape varies with model parameters.

Effect of σ_m^2 (match heterogeneity): Higher σ_m^2 increases the variance of the gap distribution, spreading mass toward the tails. This strengthens both the 5-star peak (because more positive gaps are very large) and the 1-star tail (because more negative gaps are very large). The middle bins become hollower. In the limit as $\sigma_m^2 \to \infty$, the distribution approaches uniform over bins, and the J-curve disappears.

Effect of σ_η^2 (mispricing variance): Higher σ_η^2 also increases the variance of the gap distribution at market entry. The effects are similar to higher σ_m^2: stronger tails, hollower middle. However, σ_η^2 also affects the distribution of mispricing across products, changing the composition of lemons versus bargains in the market.

Effect of c (disappointment threshold): Higher c widens the negative bins, moving mass from bin 1 to bins 2–4. This reduces the prominence of the 1-star tail and fills in the middle. Lower c has the opposite effect: a sharper 1-star spike and emptier middle bins.

Effect of β_-/β_+ (loss aversion ratio): Higher loss aversion amplifies the 1-star tail relative to the 5-star peak. As $\beta_-/\beta_+ \to 1$ (no loss aversion), the posting probabilities become symmetric, and the J-curve loses its asymmetric tail. As $\beta_-/\beta_+ \to \infty$ (extreme loss aversion), negative gaps dominate posting, and the distribution shifts toward lower ratings.

Effect of μ_d (mean posting cost): Higher μ_d reduces overall posting probability, especially for

moderate experiences. This hollows the middle and sharpens the tails (both 5-star and 1-star), making the J-curve more pronounced. Lower μ_d increases posting across all experience levels, filling in the middle and reducing the J-curve prominence.